

KETRZYN, Poland

WMO: 121850

Lat: 54.0672N

Lon: 21.3668E

Elev: 361

StdP: 14.51

Time Zone: 1.00 (EUC)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	2.0	7.7	-2.9	4.8	3.1	2.2	6.2	9.1	26.1	31.3	23.5	31.6	4.6	100	0.569

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.2	83.7	69.0	80.4	66.7	77.6	65.3	71.3	80.3	69.2	77.1	67.1	74.6	10.4	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.2	104.9	76.3	66.2	97.8	73.6	64.3	91.5	70.9	35.3	80.3	33.5	77.3	31.8	74.8	79.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.6	18.2	16.3	DB	-3.7	89.5	7.3	3.0	-8.9	91.7	-13.2	93.4	-17.3	95.1	-22.6	97.3	(4)
			WB	-4.2	74.8	7.1	2.0	-9.3	76.2	-13.5	77.4	-17.5	78.5	-22.6	79.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	46.9	27.0	28.3	35.9	47.0	55.6	61.0	65.5	64.6	57.3	47.4	39.3	32.8
	DBStd	15.33	10.39	9.74	8.44	7.38	7.46	6.17	5.68	5.34	5.60	6.42	6.40	8.21
	HDD50	2922	714	609	440	141	28	1	0	0	8	127	321	533
	HDD65	6796	1178	1029	904	540	305	151	64	74	239	544	770	998
	CDD50	1796	0	0	2	52	201	330	481	454	227	48	1	0
	CDD65	194	0	0	0	1	14	30	79	63	7	0	0	0
	CDH74	1577	0	0	0	20	130	285	637	452	53	0	0	0
	CDH80	321	0	0	0	2	11	51	156	96	5	0	0	0
Wind	WSAvg	8.0	9.3	8.5	8.3	7.7	7.2	7.1	6.7	6.8	7.2	8.4	9.3	9.5
Precipitation	PrecAvg	24.6	1.5	1.3	1.5	1.4	2.2	2.9	3.3	3.0	2.2	2.0	2.0	1.5
	PrecMax	33.6	3.4	2.3	3.7	3.7	4.7	5.7	6.7	7.8	5.0	5.6	4.1	2.8
	PrecMin	18.1	0.4	0.4	0.3	0.1	1.2	0.7	0.9	0.6	0.9	0.1	0.6	0.4
	PrecStd	3.2	0.7	0.5	0.8	0.8	0.7	1.1	1.4	1.5	1.1	1.3	0.8	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.5	48.0	60.0	77.0	81.6	85.8	88.9	87.2	79.7	67.7	54.7	50.2
		MCWB	45.7	44.9	49.9	60.8	65.5	70.8	70.9	71.6	66.1	58.8	51.7	48.0
	2%	DB	44.3	44.0	55.0	69.9	78.2	80.6	84.3	82.7	74.6	63.2	51.4	47.2
		MCWB	42.7	41.0	46.8	55.7	64.1	66.5	69.4	69.0	62.9	56.8	49.3	45.2
	5%	DB	42.0	41.4	50.6	65.0	74.0	77.2	80.6	78.9	70.8	59.8	49.4	44.7
		MCWB	40.4	39.2	44.9	52.1	60.8	64.6	67.1	66.5	60.8	55.1	47.4	42.9
	10%	DB	39.5	39.0	47.0	60.5	69.9	73.8	77.3	75.6	67.5	56.9	47.6	42.5
		MCWB	38.1	37.3	42.5	49.6	58.9	62.8	65.5	64.8	59.3	53.5	45.9	41.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.0	44.7	52.3	61.2	68.6	72.9	73.8	73.4	67.9	60.9	52.3	48.4
		MCDB	47.9	47.4	57.8	75.3	78.1	83.3	84.3	83.9	76.6	66.0	53.9	49.9
	2%	WB	43.0	41.4	48.0	56.4	64.9	69.0	71.6	70.6	64.5	57.8	49.7	45.6
		MCDB	44.2	43.5	52.9	67.0	74.5	77.4	80.9	78.9	72.1	61.5	51.1	47.0
	5%	WB	40.4	39.4	45.4	53.5	62.7	66.3	69.7	68.3	62.1	55.7	47.8	43.2
		MCDB	41.8	41.3	49.5	63.1	72.1	73.6	77.7	75.9	68.1	58.8	49.1	44.3
	10%	WB	38.1	37.4	43.0	50.9	60.2	64.1	67.4	66.3	60.3	53.9	46.2	41.2
		MCDB	39.3	39.0	46.6	58.6	67.9	71.6	74.6	73.3	66.0	56.7	47.5	42.5
Mean Daily Temperature Range	5% DB	MDBR	7.7	8.7	12.3	17.7	18.5	17.9	17.2	16.9	15.2	11.8	7.1	6.5
		MCDBR	7.6	9.8	17.8	25.0	23.4	23.0	21.8	21.3	21.0	15.8	9.3	7.7
		MCWBR	7.0	8.0	11.5	13.4	11.6	11.6	9.0	9.9	10.5	10.0	7.7	7.2
	5% WB	MCDBR	7.5	9.6	16.0	22.5	20.8	20.4	19.3	19.0	17.8	14.0	8.4	7.5
MCWBR		7.0	8.0	11.2	13.0	11.5	11.7	9.5	9.7	9.8	9.6	7.4	7.1	
Clear-Sky Solar Irradiance	taub	0.300	0.331	0.366	0.408	0.397	0.395	0.428	0.420	0.378	0.357	0.342	0.299	
	taud	2.218	2.220	2.238	2.200	2.290	2.323	2.251	2.274	2.366	2.409	2.300	2.248	
	Ebn at Noon	203	236	255	259	269	271	258	253	249	228	186	178	
	Edh at Noon	20	28	34	40	39	38	40	37	30	23	19	16	
All-Sky Solar Radiation	RadAvg	187	416	807	1308	1689	1753	1617	1382	984	509	192	126	
	RadStd	31	70	125	172	161	152	181	133	123	86	22	14	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (21 neighbors)		+0.77	N/A	N/A	N/A	N/A	N/A	N/A	-254	+126	N/A

Nomenclature: See separate page